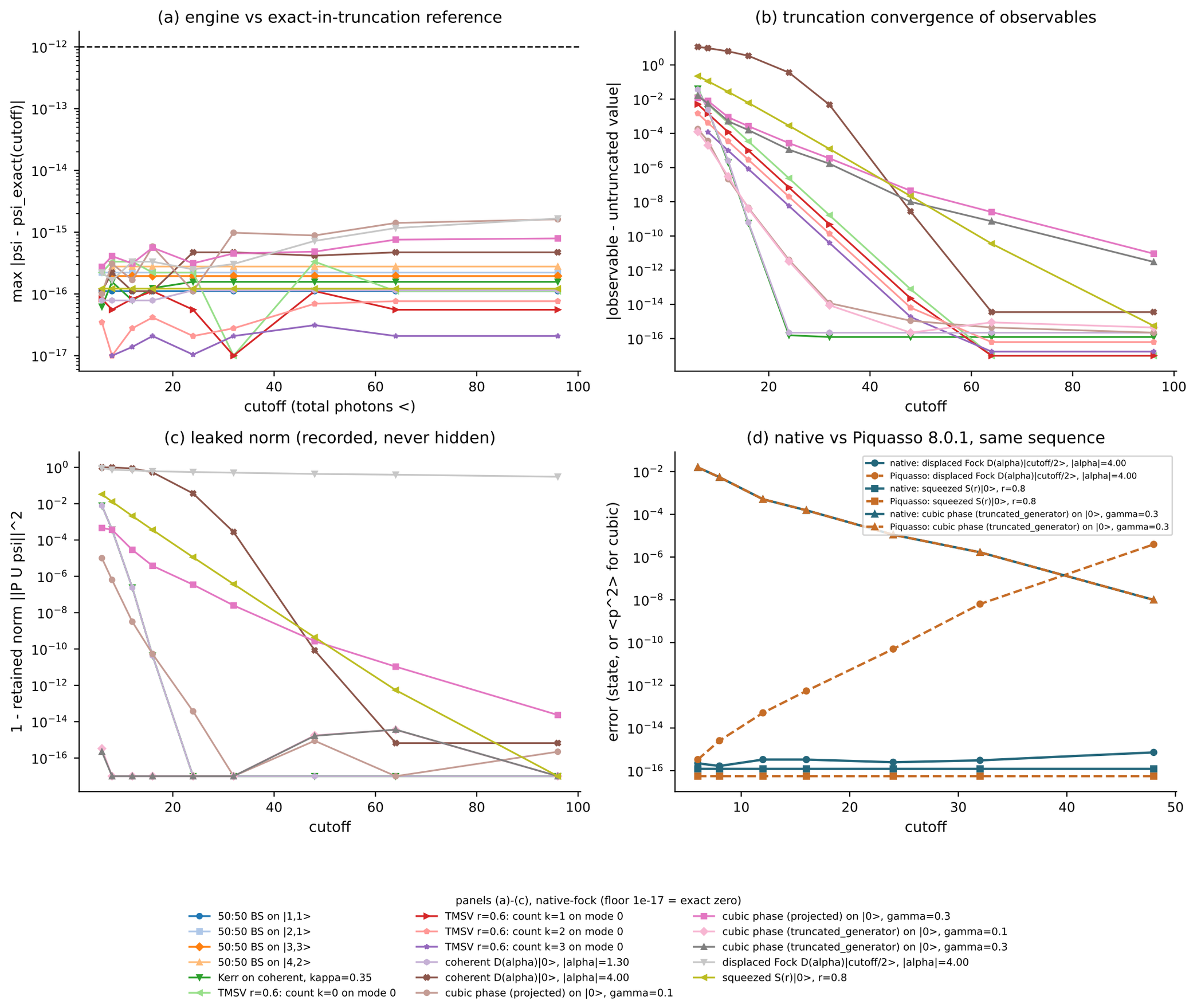
(a) engine vs exact-in-truncation reference
max |psi - psi_exact(cutoff)|
cutoff (total photons <)
(b) truncation convergence of observables
|observable - untruncated value|
cutoff
(c) leaked norm (recorded, never hidden)
1 - retained norm ||P U psi||^2
cutoff
(d) native vs Piquasso 8.0.1, same sequence
error (state, or <p^2> for cubic)
cutoff
native: displaced Fock D(alpha)|cutoff/2>, |alpha|=4.00
Piquasso: displaced Fock D(alpha)|cutoff/2>, |alpha|=4.00
native: squeezed S(r)|0>, r=0.8
Piquasso: squeezed S(r)|0>, r=0.8
native: cubic phase (truncated_generator) on |0>, gamma=0.3
Piquasso: cubic phase (truncated_generator) on |0>, gamma=0.3
panels (a)-(c), native-fock (floor 1e-17 = exact zero)
50:50 BS on |1,1>
50:50 BS on |2,1>
50:50 BS on |3,3>
50:50 BS on |4,2>
Kerr on coherent, kappa=0.35
TMSV r=0.6: count k=0 on mode 0
TMSV r=0.6: count k=1 on mode 0
TMSV r=0.6: count k=2 on mode 0
TMSV r=0.6: count k=3 on mode 0
coherent D(alpha)|0>, |alpha|=1.30
coherent D(alpha)|0>, |alpha|=4.00
cubic phase (projected) on |0>, gamma=0.1
cubic phase (projected) on |0>, gamma=0.3
cubic phase (truncated_generator) on |0>, gamma=0.1
cubic phase (truncated_generator) on |0>, gamma=0.3
displaced Fock D(alpha)|cutoff/2>, |alpha|=4.00
squeezed S(r)|0>, r=0.8